

```
# Install spaCy
!pip install spacy

# Download English language model
!python -m spacy download en_core_web_sm
```

```
ly satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
ly satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-pa
ly satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-pa
ly satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-pack
ly satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (
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ly satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages
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ly satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages
ly satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packa
ly satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (
ly satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from s
ly satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packag
ly satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.1
ly satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (
ly satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spac
ly satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from
ly satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-package
ly satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages
ly satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-pack
ly satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packa
ly satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packa
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ly satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (f
ly satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (f
ly satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (f
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ly satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from t
ly satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-pac
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ly satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from ty
ly satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages
ly satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
ly satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packag
ly satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from mark
:-web-sm==3.8.0
```

https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/
 12.8/12.8 MB 79.2 MB/s eta 0:00:00

Installation successful

Load the package via `spacy.load('en_core_web_sm')`

Read dependencies

In Jupyter or Colab notebook, you may need to restart Python in the package's dependencies. You can do this by selecting the 'Restart runtime' option.

```
import spacy

# Load English model
nlp = spacy.load("en_core_web_sm")

# Verify model loading
print("spaCy model loaded successfully!")
```

spaCy model loaded successfully!

```
sentences = [
    "The dog barked loudly.",
    "She reads a book every night.",
    "The teacher explained the lesson clearly.",
    "The cat sat on the mat.",
    "He plays football and watches movies.",
    "The company launched a new product yesterday.",
    "If it rains, we will cancel the trip.",
    "The scientist who discovered the vaccine won an award.",
    "Why did you open the door?",
    "Where are they going now?",
    "The cake was baked by my mother.",
    "The report was written by the manager.",
    "Students are learning natural language processing.",
    "Social media influences public opinion.",
    "The software developer fixed the bug quickly."
]
```

```
docs = [nlp(sentence) for sentence in sentences]

for doc in docs:
    print("Sentence:", doc.text)
    for token in doc:
        print(token.text, token.pos_, token.dep_)
    print("\n")
```

report NOUN nsubjpass
was AUX auxpass
written VERB ROOT
by ADP agent
the DET det
manager NOUN pobj
. PUNCT punct

Sentence: Students are learning natural language processing.

Students NOUN nsubj
are AUX aux
learning VERB ROOT
natural ADJ amod
language NOUN compound
processing NOUN dobj
. PUNCT punct

Sentence: Social media influences public opinion.

Social ADJ amod
media NOUN nsubj
influences NOUN ROOT
public ADJ amod
opinion NOUN dobj
. PUNCT punct

Sentence: The software developer fixed the bug quickly.

The DET det
software NOUN compound
developer NOUN nsubj
fixed VERB ROOT
the DET det
bug NOUN dobj
quickly ADV advmod
. PUNCT punct

```
import pandas as pd

data = []

for doc in docs:
    for token in doc:
        data.append({
            "Sentence": doc.text,
            "Token": token.text,
            "Lemma": token.lemma_,
            "POS": token.pos_,
            "Dependency": token.dep_,
            "Head Word": token.head.text
        })

df = pd.DataFrame(data)

df.head(20)
```

	Sentence	Token	Lemma	POS	Dependency	Head Word
0	The dog barked loudly.	The	the	DET	det	dog
1	The dog barked loudly.	dog	dog	NOUN	nsubj	barked
2	The dog barked loudly.	barked	bark	VERB	ROOT	barked
3	The dog barked loudly.	loudly	loudly	ADV	advmod	barked
4	The dog barked loudly.	.	.	PUNCT	punct	barked
5	She reads a book every night.	She	she	PRON	nsubj	reads
6	She reads a book every night.	reads	read	VERB	ROOT	reads
7	She reads a book every night.	a	a	DET	det	book
8	She reads a book every night.	book	book	NOUN	dobj	reads
9	She reads a book every night.	every	every	DET	det	night
10	She reads a book every night.	night	night	NOUN	npadvmod	reads
11	She reads a book every night.	.	.	PUNCT	punct	reads
12	The teacher explained the lesson	The	the	DET	det	teacher

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
for doc in docs:
```

```
    subject = ""
```

```
    verb = ""
```

```
    obj = ""
```

```
    for token in doc:
```

```
        if token.dep_ == "nsubj":
```

```
            subject = token.text
```

```
        if token.dep_ == "ROOT":
```

```
            verb = token.text
```

```
        if token.dep_ in ("dobj", "pobj"):
```

```
            obj = token.text
```

```
    print("Sentence:", doc.text)
```

```
    print("Subject:", subject)
```

```
    print("Verb:", verb)
```

```
    print("Object:", obj)
```

```
    print("-----")
```

```
Verb: sat
```

```
Object: mat
```

```
-----
```

```
Sentence: He plays football and watches movies.
```

```
Subject: He
```

```
Verb: plays
```

```
Object: movies
```

```
-----
```

```
Sentence: The company launched a new product yesterday.
```

```
Subject: company
```

```
Verb: launched
```

```
Object: product
```

```
-----
```

```
Sentence: If it rains, we will cancel the trip.
```

```
Subject: we
```

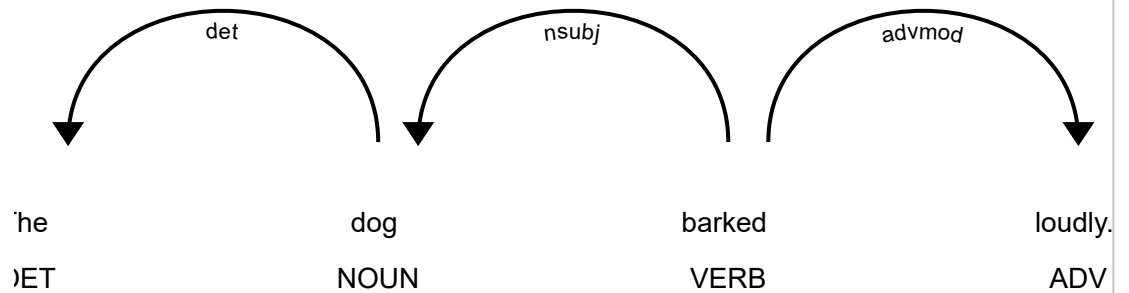
```
Verb: cancel
```

```
Object: trip
```

```
Subject:
Verb: baked
Object: mother
-----
Sentence: The report was written by the manager.
Subject:
Verb: written
Object: manager
-----
Sentence: Students are learning natural language processing.
Subject: Students
Verb: learning
Object: processing
-----
Sentence: Social media influences public opinion.
Subject: media
Verb: influences
Object: opinion
-----
Sentence: The software developer fixed the bug quickly.
Subject: developer
Verb: fixed
Object: bug
-----
```

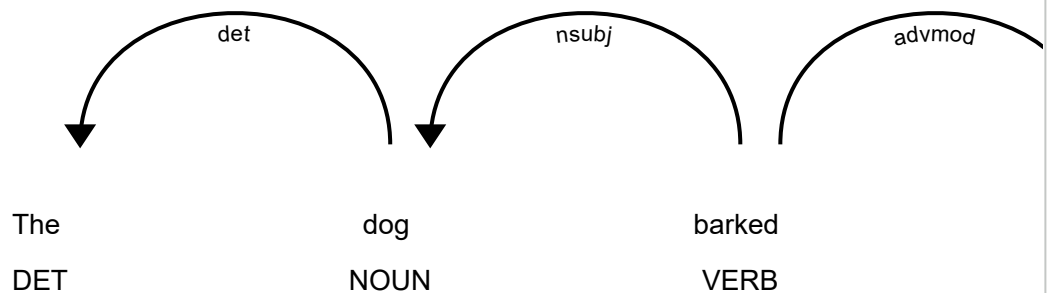
```
from spacy import displacy
```

```
for i in range(5):
    doc = nlp(sentences[i])
    displacy.render(doc, style="dep", jupyter=True)
```

```
displacy.serve(nlp(sentences[0]), style="dep")
```

```
/usr/local/lib/python3.12/dist-packages/spacy/displacy/_init_.py:108: UserWarning:
  warnings.warn(Warnings.W011)
```



Using the 'dep' visualizer
Serving on <http://0.0.0.0:5000> ...

dobj